

SULIT



SECP2523 DATABASE

SESSION 2024/2025 - SEMESTER 1

ALTERNATIVE ASSESSMENT REPORT: PHASE 2

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SECTION C (SQL AND INTERFACE IMPLEMENTATION)

3.1 SQL Statement (DDL & DML)

- a) DDL statement
 - DDL of Membership table

```
1 CREATE TABLE Membership (  
2     applicationID INT(11) NOT NULL PRIMARY KEY,  
3     email VARCHAR(100) UNIQUE NOT NULL,  
4     name VARCHAR(100) NOT NULL,  
5     icNo VARCHAR(14) UNIQUE NOT NULL,  
6     marriageStatus ENUM('Single', 'Married', 'Divorced', 'Widowed') NOT NULL,  
7     homeAddress TEXT NOT NULL,  
8     homePostcode INT(5) NOT NULL,  
9     homeState VARCHAR(50) NOT NULL,  
10    homeCity VARCHAR(50) NOT NULL,  
11    gender ENUM('Male', 'Female') NOT NULL,  
12    religion VARCHAR(20) NOT NULL,  
13    race VARCHAR(20) NOT NULL,  
14    monthlySalary DECIMAL(10, 2) NOT NULL,  
15    staffNo VARCHAR(50) NOT NULL,  
16    pfNo VARCHAR(50) NOT NULL,  
17    positionGrade VARCHAR(20) NOT NULL,  
18    workAddress TEXT NOT NULL,  
19    workState VARCHAR(50) NOT NULL,  
20    workCity VARCHAR(50) NOT NULL,  
21    workPostcode INT(5) NOT NULL,  
22    faxNo INT(15),  
23    phoneNo INT(15) UNIQUE NOT NULL,  
24    homePhoneNo INT(15),  
25    entranceFee DECIMAL(10, 2) NOT NULL,  
26    shareCapital DECIMAL(10, 2) NOT NULL,  
27    feeCapital DECIMAL(10, 2) NOT NULL,  
28    memberDeposit DECIMAL(10, 2) NOT NULL,  
29    charityFund DECIMAL(10, 2) NOT NULL,  
30    fixedSavings DECIMAL(10, 2) NOT NULL,  
31    applicationDate DATE NOT NULL,  
32    applicationStatus ENUM('Success', 'Pending', 'Rejected') NOT NULL,  
33    membershipStatus ENUM('Active', 'Inactive') NOT NULL,  
34    approvalDate DATE,  
35    comment TEXT  
36 );
```

Figure 3.1.1 DDL of Membership table

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0006 seconds.)

```
CREATE TABLE Membership ( applicationID INT(11) NOT NULL PRIMARY KEY, email
VARCHAR(100) UNIQUE NOT NULL, name VARCHAR(100) NOT NULL, icNo VARCHAR(14) UNIQUE
NOT NULL, marriageStatus ENUM('Single', 'Married', 'Divorced', 'Widowed') NOT NULL,
homeAddress TEXT NOT NULL, homePostcode INT(5) NOT NULL, homeState VARCHAR(50) NOT
NULL, homeCity VARCHAR(50) NOT NULL, gender ENUM('Male', 'Female') NOT NULL,
religion VARCHAR(20) NOT NULL, race VARCHAR(20) NOT NULL, monthlySalary DECIMAL(10,
2) NOT NULL, staffNo VARCHAR(50) NOT NULL, pfNo VARCHAR(50) NOT NULL, positionGrade
VARCHAR(20) NOT NULL, workAddress TEXT NOT NULL, workState VARCHAR(50) NOT NULL,
workCity VARCHAR(50) NOT NULL, workPostcode INT(5) NOT NULL, faxNo INT(15), phoneNo
INT(15) UNIQUE NOT NULL, homePhoneNo INT(15), entranceFee DECIMAL(10, 2) NOT NULL,
shareCapital DECIMAL(10,[...])
```

[Edit]

Figure 3.1.2 Result from DDL of Membership table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	applicationID 	int(11)			No	None		
2	email 	varchar(100)	utf8mb4_general_ci		No	None		
3	name	varchar(100)	utf8mb4_general_ci		No	None		
4	icNo 	varchar(14)	utf8mb4_general_ci		No	None		
5	marriageStatus	enum('Single', 'Married', 'Divorced', 'Widowed')	utf8mb4_general_ci		No	None		
6	homeAddress	text	utf8mb4_general_ci		No	None		
7	homePostcode	int(5)			No	None		
8	homeState	varchar(50)	utf8mb4_general_ci		No	None		
9	homeCity	varchar(50)	utf8mb4_general_ci		No	None		
10	gender	enum('Male', 'Female')	utf8mb4_general_ci		No	None		
11	religion	varchar(20)	utf8mb4_general_ci		No	None		
12	race	varchar(20)	utf8mb4_general_ci		No	None		
13	monthlySalary	decimal(10,2)			No	None		
14	staffNo	varchar(50)	utf8mb4_general_ci		No	None		
15	pfNo	varchar(50)	utf8mb4_general_ci		No	None		
16	positionGrade	varchar(20)	utf8mb4_general_ci		No	None		
17	workAddress	text	utf8mb4_general_ci		No	None		
18	workState	varchar(50)	utf8mb4_general_ci		No	None		

19	workCity	varchar(50)	utf8mb4_general_ci	No	None
20	workPostcode	int(5)		No	None
21	faxNo	int(15)		Yes	NULL
22	phoneNo 	int(15)		No	None
23	homePhoneNo	int(15)		Yes	NULL
24	entranceFee	decimal(10,2)		No	None
25	shareCapital	decimal(10,2)		No	None
26	feeCapital	decimal(10,2)		No	None
27	memberDeposit	decimal(10,2)		No	None
28	charityFund	decimal(10,2)		No	None
29	fixed Savings	decimal(10,2)		No	None
30	applicationDate	date		No	None
31	application Status	enum('Success', 'Pending', 'Rejected')	utf8mb4_general_ci	No	None
32	membershipStatus	enum('Active', 'Inactive')	utf8mb4_general_ci	No	None
33	approvalDate	date		Yes	NULL
34	comment	text	utf8mb4_general_ci	Yes	NULL

Figure 3.1.3 Membership table structure

- DDL of KADAAAdmin table

```

1 CREATE TABLE KADAAAdmin (
2     adminID INT(11) PRIMARY KEY NOT NULL
3 );

```

Figure 3.1.4 DDL of KADAAAdmin table

 MySQL returned an empty result set (i.e. zero rows). (Query took 0.0009 seconds.)

```
CREATE TABLE KADAAAdmin ( adminID INT(11) PRIMARY KEY NOT NULL );
```

[\[Edit inline \]](#) [\[Edit \]](#) [\[Create PHP code \]](#)

Figure 3.1.5 Result from DDL of KADAAAdmin table

Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
adminID 	int(11)			No	None			 Change  Drop

Figure 3.1.6 KADAAAdmin table structure

- DDL of BoardOfDirector table

```
1 CREATE TABLE BoardOfDirector (
2     bodID INT(11) PRIMARY KEY NOT NULL
3 );
```

Figure 3.1.7 DDL of BoardOfDirector table

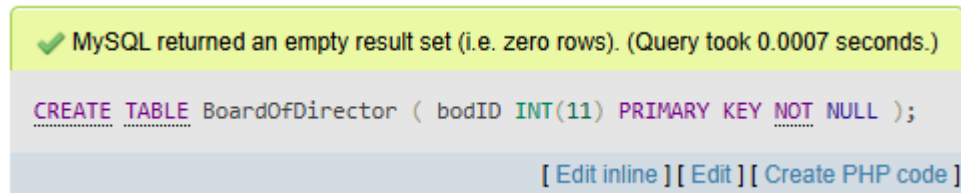


Figure 3.1.8 Result from DDL of BoardOfDirector table

Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
bodID	int(11)			No	None			Change Drop More

Figure 3.1.9 BoardOfDirector table structure

- DDL of Loan table

```
1 CREATE TABLE Loan (
2     loanApplicationID INT(11) PRIMARY KEY NOT NULL,
3     applicationID INT(11) NOT NULL,
4     loanType ENUM('Al-Bai', 'Al-Innah', 'Baik Pulih Kendaraan', 'Road Tax dan
5     Insurans', 'Khas', 'Karnival Musim Istimewa') NOT NULL,
6     loanStatus ENUM('Accepted', 'Pending', 'Rejected') NOT NULL,
7     loanAmount DECIMAL(10, 2) NOT NULL,
8     loanPeriod INT(3) NOT NULL,
9     monthlyInstallment DECIMAL(10, 2),
10    DOB DATE NOT NULL,
11    age INT(2) NOT NULL,
12    bankName VARCHAR(100) NOT NULL,
13    bankAccountNo VARCHAR(30) NOT NULL,
14    guarantorName VARCHAR(100) NOT NULL,
15    guarantorICNo VARCHAR(14) NOT NULL,
16    guarantorPFNo VARCHAR(20) NOT NULL,
17    guarantorMemberID INT(11) NOT NULL,
18    guarantorPhoneNo INT(15) NOT NULL,
19    loanPaymentReceipt TEXT NOT NULL,
20    FOREIGN KEY (applicationID) REFERENCES Membership(applicationID)
21 );
```

Figure 3.1.10 DDL of Loan table

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0005 seconds.)

```
CREATE TABLE Loan ( loanApplicationID INT(11) PRIMARY KEY NOT NULL, applicationID INT(11) NOT NULL, loanType ENUM('Al-Bai', 'Al-Innah', 'Baik Pulih Kendaraan', 'Road Tax dan Insurans', 'Khas', 'Karnival Musim Istimewa', 'Al-Qadrul Hassan') NOT NULL, loanAmount DECIMAL(10, 2) NOT NULL, loanPeriod INT(3) NOT NULL, monthlyInstallment DECIMAL(10, 2), DOB DATE NOT NULL, age INT(2) NOT NULL, bankName VARCHAR(100) NOT NULL, bankAccountNo VARCHAR(30) NOT NULL, guarantorName VARCHAR(100) NOT NULL, guarantorICNo VARCHAR(14) NOT NULL, guarantorPFNo VARCHAR(20) NOT NULL, guarantorMemberID INT(11) NOT NULL, guarantorPhoneNo INT(15) NOT NULL, loanPaymentReceipt TEXT NOT NULL, FOREIGN KEY (applicationID) REFERENCES Membership(applicationID) );
```

[Edit inline] [Edit] [Create PHP code]

Figure 3.1.11 Result from DDL of Loan table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	loanApplicationID	int(11)			No	None		
2	applicationID	int(11)			No	None		
3	loanType	enum('Al-Bai', 'Al-Innah', 'Baik Pulih Kendaraan', 'Road Tax dan Insurans', 'Khas', 'Karnival Musim Istimewa', 'Al-Qadrul Hassan')	utf8mb4_general_ci		No	None		
4	loanAmount	decimal(10,2)			No	None		
5	loanPeriod	int(3)			No	None		
6	monthlyInstallment	decimal(10,2)			Yes	NULL		
7	DOB	date			No	None		
8	age	int(2)			No	None		
9	bankName	varchar(100)	utf8mb4_general_ci		No	None		
10	bankAccountNo	varchar(30)	utf8mb4_general_ci		No	None		
11	guarantorName	varchar(100)	utf8mb4_general_ci		No	None		
12	guarantorICNo	varchar(14)	utf8mb4_general_ci		No	None		
13	guarantorPFNo	varchar(20)	utf8mb4_general_ci		No	None		
14	guarantorMemberID	int(11)			No	None		
15	guarantorPhoneNo	int(15)			No	None		
16	loanPaymentReceipt	text	utf8mb4_general_ci		No	None		
17	loanStatus	enum('Accepted', 'Pending', 'Rejected')	utf8mb4_general_ci		No	Pending		

Figure 3.1.12 Loan table structure

- DDL of Stock table

```

1 CREATE TABLE Stock (
2     memberNo INT(11) NOT NULL,
3     recordDate DATE NOT NULL,
4     applicationID INT(11) NOT NULL,
5     shareCapital DECIMAL(12, 2) DEFAULT 0.00,
6     feeCapital DECIMAL(12, 2) DEFAULT 0.00,
7     fixedSavings DECIMAL(12, 2) DEFAULT 0.00,
8     memberSavings DECIMAL(12, 2) DEFAULT 0.00,
9     memberFund DECIMAL(12, 2) DEFAULT 0.00,
10    PRIMARY KEY (memberNo, recordDate),
11    FOREIGN KEY (applicationID) REFERENCES membership(applicationID)
12 );

```

Figure 3.1.13 DDL of Stock table

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0006 seconds.)

```

CREATE TABLE Stock ( memberNo INT(11) NOT NULL, recordDate DATE NOT NULL, applicationID INT(11)
NOT NULL, shareCapital DECIMAL(12, 2) DEFAULT 0.00, feeCapital DECIMAL(12, 2) DEFAULT 0.00,
fixedSavings DECIMAL(12, 2) DEFAULT 0.00, memberSavings DECIMAL(12, 2) DEFAULT 0.00, memberFund
DECIMAL(12, 2) DEFAULT 0.00, PRIMARY KEY (memberNo, recordDate), FOREIGN KEY (applicationID)
REFERENCES membership(applicationID) );

```

[\[Edit inline \]](#) [\[Edit \]](#) [\[Create PHP code \]](#)

Figure 3.1.14 Result from DDL of Stock table

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	memberNo 🔑	int(11)			No	None		
2	recordDate 🔑	date			No	None		
3	applicationID 🔑	int(11)			No	None		
4	shareCapital	decimal(12,2)			Yes	0.00		
5	feeCapital	decimal(12,2)			Yes	0.00		
6	fixedSavings	decimal(12,2)			Yes	0.00		
7	memberSavings	decimal(12,2)			Yes	0.00		
8	memberFund	decimal(12,2)			Yes	0.00		

Figure 3.1.14 Stock table structure

- DDL of ReportHub table

```

1 CREATE TABLE ReportHub (
2     Month INT(2) NOT NULL,
3     Year INT(4) NOT NULL,
4     adminID INT(11) NOT NULL,
5     bodID INT(11),
6     memberNo INT(11),
7     total_membershipApplicationStatus INT DEFAULT NULL,
8     total_loanApplicationStatus INT DEFAULT NULL,
9
10    total_shareCapital DECIMAL(12, 2) DEFAULT 0.00,
11    total_feeCapital DECIMAL(12, 2) DEFAULT 0.00,
12    total_membersFund DECIMAL(12, 2) DEFAULT 0.00,
13    PRIMARY KEY (Month, Year),
14    FOREIGN KEY (adminID) REFERENCES kadaadmin(adminID),
15    FOREIGN KEY (bodID) REFERENCES boardofdirector(bodID),
16    FOREIGN KEY (memberNo) REFERENCES stock(memberNo)
17 );

```

Figure 3.1.15 DDL of ReportHub table

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0008 seconds.)

```

CREATE TABLE ReportHub ( Month INT(2) NOT NULL, Year INT(4) NOT NULL, adminID INT(11) NOT
NULL, bodID INT(11), memberNo INT(11), total_membershipApplicationStatus INT DEFAULT NULL,
total_loanApplicationStatus INT DEFAULT NULL, total_loanTypeApplication INT DEFAULT NULL,
total_shareCapital DECIMAL(12, 2) DEFAULT 0.00, total_feeCapital DECIMAL(12, 2) DEFAULT 0.00,
total_membersFund DECIMAL(12, 2) DEFAULT 0.00, PRIMARY KEY (Month, Year), FOREIGN KEY
(adminID) REFERENCES kadaadmin(adminID), FOREIGN KEY (bodID) REFERENCES
boardofdirector(bodID), FOREIGN KEY (memberNo) REFERENCES stock(memberNo) );

```

[\[Edit inline \]](#) [\[Edit \]](#) [\[Create PHP code \]](#)

Figure 3.1.16 Result from DDL of ReportHub table

#	Name	Type	Collation	Attributes	Null	Default
1	Month 	int(2)			No	None
2	Year 	int(4)			No	None
3	adminID 	int(11)			No	None
4	bodID 	int(11)			Yes	NULL
5	memberNo 	int(11)			Yes	NULL
6	total_membershipApplicationStatus	int(11)			Yes	NULL
7	total_loanApplicationStatus	int(11)			Yes	NULL
9	total_shareCapital	decimal(12,2)			Yes	0.00
10	total_feeCapital	decimal(12,2)			Yes	0.00
11	total_membersFund	decimal(12,2)			Yes	0.00

Figure 3.1.17 ReportHub table structure

```

1 ALTER TABLE ReportHub
2 ADD COLUMN applicationID INT NOT NULL,
3 ADD COLUMN loanApplicationID INT NOT NULL,
4 ADD CONSTRAINT fk_applicationID FOREIGN KEY (applicationID) REFERENCES
  Membership(applicationID),
5 ADD CONSTRAINT fk_loanApplicationID FOREIGN KEY (loanApplicationID) REFERENCES
  Loan(loanApplicationID);

```

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0015 seconds.)

```

ALTER TABLE ReportHub ADD COLUMN applicationID INT NOT NULL, ADD COLUMN loanApplicationID INT NOT
NULL, ADD CONSTRAINT fk_applicationID FOREIGN KEY (applicationID) REFERENCES
Membership(applicationID), ADD CONSTRAINT fk_loanApplicationID FOREIGN KEY (loanApplicationID)
REFERENCES Loan(loanApplicationID);

```

[\[Edit inline \]](#) [\[Edit \]](#) [\[Create PHP code \]](#)

Figure 3.1.18 Adding applicationID and loanApplicationID into ReportHub table

b) DML of INSERT statements

- insert data into Membership table

```

1 INSERT INTO membership (
2     applicationID, email, name, icNo, marriageStatus, homeAddress, homePostcode,
3     homeState, homeCity, gender, religion, race, monthlySalary, staffNo, pfNo,
4     positionGrade, workAddress, workState, workCity, workPostcode, faxNo,
5     phoneNo, homePhoneNo, entranceFee, shareCapital, feeCapital, memberDeposit,
6     charityFund, fixedSavings, applicationDate, applicationStatus,
7     membershipStatus, approvalDate, comment
8 )
9 VALUES
10 (3001, 'arjuna.nirmala@example.com', 'Arjuna Nirmala', '910101145678', 'Married',
11  '123 Jalan Kerinchi', '43000', 'Selangor', 'Kajang', 'Male', 'Islam', 'Malay',
12  5000, 'S12345', 'PF67890', 'Grade 1', '456 Corporate Blvd', 'Selangor',
13  'Shah Alam', '40000', '031234567', '0123456789', '039876543', 50.00, 2000.00,
14  500.00, 300.00, 100.00, 1000.00, '2025-01-01', 'Pending', 'Active',
15  '2025-01-15', 'Approved without issues'),
16
17 (3002, 'kasih.cendana@example.com', 'Kasih Cendana', '920202063456', 'Single',
18  '45 Bunga Harum', '10460', 'Penang', 'Georgetown', 'Female', 'Christianity',
19  'Chinese', 3500, 'S67890', 'PF12345', 'Grade 2', '789 Tech Park', 'Penang',
20  'Butterworth', '12200', '046789100', '0174567890', '045566778', 70.00, 1500.00,
21  300.00, 200.00, 80.00, 800.00, '2024-12-15', 'Approved', 'Active',
22  '2024-12-20', 'Application approved quickly'),

```

```

24 (3003, 'aryan.hadi@example.com', 'Aryan Hadi', '930303127890', 'Married',
25 '678 Damansara Hill', '50700', 'Kuala Lumpur', 'Kuala Lumpur', 'Male', 'Islam',
26 'Malay', 4000, 'S23456', 'PF45678', 'Grade 1', '12 Office Ave', 'Kuala Lumpur',
27 'Kuala Lumpur', '50700', '037654321', '0112233445', '038877665', 60.00, 2500.00,
28 600.00, 500.00, 120.00, 1500.00, '2025-02-10', 'Pending', 'Inactive',
29 NULL, 'Awaiting further documents'),
30
31 (3004, 'beatrice.rini@example.com', 'Beatrice Rini', '940404085678', 'Divorced',
32 '90 Riverfront', '80000', 'Johor', 'Johor Bahru', 'Female', 'Hinduism', 'Indian',
33 4500, 'S34567', 'PF67890', 'Grade 3', '34 Business Park', 'Johor',
34 'Pasir Gudang', '81700', '072233445', '0133344556', '071122334', 80.00, 3000.00,
35 700.00, 400.00, 200.00, 2000.00, '2023-11-05', 'Rejected', 'Inactive',
36 '2023-11-15', 'Application rejected due to missing documents');

```

Figure 3.1.18 Inserting data into Membership table

✔ 4 rows inserted. (Query took 0.0037 seconds.)

```

INSERT INTO membership ( applicationID, email, name, icNo, marriageStatus, homeAddress,
homePostcode, homeState, homeCity, gender, religion, race, monthlySalary, staffNo, pfNo,
positionGrade, workAddress, workState, workCity, workPostcode, faxNo, phoneNo, homePhoneNo,
entranceFee, shareCapital, feeCapital, memberDeposit, charityFund, fixedSavings,
applicationDate, applicationStatus, membershipStatus, approvalDate, comment ) VALUES (3001,
'arjuna.nirmala@example.com', 'Arjuna Nirmala', '910101145678', 'Married', '123 Jalan
Kerinci', '43000', 'Selangor', 'Kajang', 'Male', 'Islam', 'Malay', 5000, 'S12345', 'PF67890',
'Grade 1', '456 Corporate Blvd', 'Selangor', 'Shah Alam', '40000', '031234567', '0123456789',
'039876543', 50.00, 2000.00, 500.00, 300.00, 100.00, 1000.00, '2025-01-01', 'Approved',
'Active', '2025-01-15', 'Approved without issues'), (3002, 'kasih.cendana@example.com', 'Kasih
Cendana', '920202063456', 'Single', '45[...]

```

[\[Edit \]](#)

Figure 3.1.19 Result after inserting data into Membership table

applicationID	email	name	icNo	marriageStatus	homeAddress	homePostcode	homeState	homeCity	gender	religion	race	monthlySalary	staffNo	pfNo	positionGrade
3001	arjuna.nirmala@example.com	Arjuna Nirmala	910101145678	Married	123 Jalan Kerinci	43000	Selangor	Kajang	Male	Islam	Malay	5000.00	S12345	PF67890	Grade 1
3002	kasih.cendana@example.com	Kasih Cendana	920202063456	Single	45 Bunga Harum	10460	Penang	Georgetown	Female	Christianity	Chinese	3500.00	S67890	PF12345	Grade 2
3003	aryan.hadi@example.com	Aryan Hadi	930303127890	Married	678 Damansara Hill	50700	Kuala Lumpur	Kuala Lumpur	Male	Islam	Malay	4000.00	S23456	PF45678	Grade 1
3004	beatrice.rini@example.com	Beatrice Rini	940404085678	Divorced	90 Riverfront	80000	Johor	Johor Bahru	Female	Hinduism	Indian	4500.00	S34567	PF67890	Grade 3

workAddress	workState	workCity	workPostcode	faxNo	phoneNo	homePhoneNo	entranceFee	shareCapital	feeCapital	memberDeposit	charityFund	fixedSavings	applicationDate	applicationStatus
456 Corporate Blvd	Selangor	Shah Alam	40000	31234567	123456789	39876543	50.00	2000.00	500.00	300.00	100.00	1000.00	2025-01-01	Success
789 Tech Park	Penang	Butterworth	12200	46789100	174567890	45566778	70.00	1500.00	300.00	200.00	80.00	800.00	2024-12-15	Success
12 Office Ave	Kuala Lumpur	Kuala Lumpur	50700	37654321	112233445	38877665	60.00	2500.00	600.00	500.00	120.00	1500.00	2025-02-10	Pending
34 Business Park	Johor	Pasir Gudang	81700	72233445	133344556	71122334	80.00	3000.00	700.00	400.00	200.00	2000.00	2023-11-05	Rejected

membershipStatus	approvalDate	comment
Active	2025-01-15	Approved without issues
Active	2024-12-20	Application approved quickly
Inactive	NULL	Awaiting further documents
Inactive	2023-11-15	Application rejected due to missing documents

Figure 3.1.20 Data is inserted into Membership table

- insert data into Loan table

```

1 INSERT INTO Loan (
2     loanApplicationID, applicationID, loanType, loanAmount, loanPeriod,
3     monthlyInstallment, DOB, age, bankName, bankAccountNo,
4     guarantorICNo, guarantorName, guarantorPFNo, guarantorMemberID,
5     guarantorPhoneNo, loanPaymentReceipt
6 )
7 VALUES
8 -- Loan for Arjuna Nirmala
9 (1001, 3001, 'Al-Innah', 50000.00, 60, 1000.00, '1991-01-01', 34,
10  'Maybank', '123456789', '890101-14-5678', 'Zainul Abidin', 'PF11223', 101,
11  '0123456789', 'Receipt_1001.pdf'),
12
13 -- Loan for Kasih Cendana
14 (1002, 3002, 'Karnival Musim Istimewa', 250000.00, 240, 1500.00, '1992-02-02',
15  33,
16  'CIMB Bank', '987654321', '870202-06-1234', 'Chang Wei', 'PF33445', 102,
17  '0174567890', 'Receipt_1002.pdf'),
18
19 -- Loan for Aryan Hadi
20 (1003, 3003, 'Al-Qadrul Hassan', 70000.00, 84, 1200.00, '1993-03-03', 31,
21  'Public Bank', '112233445', '860303-12-4567', 'Rahmat Amin', 'PF55678', 103,
22  '0112233445', 'Receipt_1003.pdf'),
23
24 -- Loan for Beatrice Rini
25 (1004, 3004, 'Baik Pulih Kendaraan', 150000.00, 120, 2500.00, '1994-04-04', 29,
26  'RHB Bank', '667788990', '880404-08-7890', 'Siti Mariam', 'PF77890', 104,
27  '0133344556', 'Receipt_1004.pdf');

```

Figure 3.1.21 Inserting data into Loan table

✓ 4 rows inserted. (Query took 0.0152 seconds.)

```

INSERT INTO Loan ( loanApplicationID, applicationID, loanType, loanAmount, loanPeriod,
monthlyInstallment, DOB, age, bankName, bankAccountNo, guarantorICNo, guarantorName,
guarantorPFNo, guarantorMemberID, guarantorPhoneNo, loanPaymentReceipt ) VALUES -- Loan for
Arjuna Nirmala (1001, 3001, 'Al-Innah', 50000.00, 60, 1000.00, '1991-01-01', 34, 'Maybank',
'123456789', '890101-14-5678', 'Zainul Abidin', 'PF11223', 101, '0123456789',
'Receipt_1001.pdf'), -- Loan for Kasih Cendana (1002, 3002, 'Karnival Musim Istimewa',
250000.00, 240, 1500.00, '1992-02-02', 33, 'CIMB Bank', '987654321', '870202-06-1234', 'Chang
Wei', 'PF33445', 102, '0174567890', 'Receipt_1002.pdf'), -- Loan for Aryan Hadi (1003, 3003,
'Al-Qadrul Hassan', 70000.00, 84, 1200.00, '1993-03-03', 31, 'Public Bank', '112233445',
'860303-12-4567', 'Rahmat Amin', 'PF55678', 103, '0112233445', 'Receipt_1003.pdf'), -- Loan
for Beatrice Rini (1004, 3004, 'Baik Pulih Kendaraan'[...]

```

[Edit]

Figure 3.1.22 Result after inserting data into Loan table

loanApplicationID	applicationID	loanType	loanAmount	loanPeriod	monthlyInstallment	DOB	age	bankName	bankAccountNo	guarantorName	guarantorICNo	guarantorPFNo	guarantorMemberID
1001	3001	Al-Innah	50000.00	60	1000.00	1991-01-01	34	Maybank	123456789	Zainul Abidin	890101-14-5678	PF11223	101
1002	3002	Karnival Musim Istimewa	250000.00	240	1500.00	1992-02-02	33	CIMB Bank	987654321	Chang Wei	870202-06-1234	PF33445	102
1003	3003	Al-Qadrul Hassan	70000.00	84	1200.00	1993-03-03	31	Public Bank	112233445	Rahmat Amin	860303-12-4567	PF55678	103
1004	3004	Baik Pulih Kendaraan	150000.00	120	2500.00	1994-04-04	29	RHB Bank	667788990	Siti Mariam	880404-08-7890	PF77890	104

guarantorPhoneNo	loanPaymentReceipt
123456789	Receipt_1001.pdf
174567890	Receipt_1002.pdf
112233445	Receipt_1003.pdf
133344556	Receipt_1004.pdf

Figure 3.1.23 Data is inserted into Loan table

- insert data into Stock table

```

1 INSERT INTO Stock (
2     memberNo, recordDate, applicationID, shareCapital, feeCapital, fixedSaving,
3     memberSavings, memberFund
4 )
5 VALUES
6 -- Stock record for Arjuna Nirmala
7 (101, '2025-01-15', 3001,
8  2000.00, 500.00, 1000.00,
9  (2000.00 + 500.00 + 1000.00),
10 (2000.00 + 500.00 + 1000.00)),
11
12 -- Stock record for Kasih Cendana
13 (102, '2024-12-20', 3002,
14  1500.00, 300.00, 800.00,
15  (1500.00 + 300.00 + 800.00 + 2000.00 + 500.00 + 1000.00),
16  (1500.00 + 300.00 + 800.00 + 2000.00 + 500.00 + 1000.00));

```

Figure 3.1.24 Inserting data into Stock table

✓ 2 rows inserted. (Query took 0.0186 seconds.)

```

INSERT INTO Stock ( memberNo, recordDate, applicationID, shareCapital, feeCapital, fixedSaving,
memberSavings, memberFund ) VALUES -- Stock record for Arjuna Nirmala (101, '2025-01-15', 3001,
2000.00, 500.00, 1000.00, (2000.00 + 500.00 + 1000.00), (2000.00 + 500.00 + 1000.00)), -- Stock
record for Kasih Cendana (102, '2024-12-20', 3002, 1500.00, 300.00, 800.00, (1500.00 + 300.00 +
800.00 + 2000.00 + 500.00 + 1000.00), (1500.00 + 300.00 + 800.00 + 2000.00 + 500.00 + 1000.00));

```

[\[Edit inline \]](#) [\[Edit \]](#) [\[Create PHP code \]](#)

Figure 3.1.25 Result after inserting data into Stock table

memberNo	recordDate	applicationID	shareCapital	feeCapital	fixedSaving	memberSavings	memberFund
101	2025-01-15	3001	2000.00	500.00	1000.00	3500.00	3500.00
102	2024-12-20	3002	1500.00	300.00	800.00	6100.00	6100.00

Figure 3.1.26 Data is inserted into Stock table

- insert data into ReportHub table

```

1 INSERT INTO ReportHub (
2     Month, Year, adminID, bodID, memberNo, applicationID, loanApplicationID,
3     total_membershipApplicationStatus, total_loanApplicationStatus,
4     total_loanTypeApplication, total_shareCapital, total_feeCapital,
5     total_membersFund
6 )
7 VALUES (
8     '01', '2025', 1, 1, 101,
9     (SELECT COUNT(*) FROM Membership),
10    (SELECT COUNT(*) FROM Loan),
11    (SELECT COUNT(*) FROM Membership WHERE membershipStatus = 'Active'),
12    (SELECT COUNT(*) FROM Loan WHERE loanApplicationStatus = 'Approved'),
13    (SELECT COUNT(*) FROM Loan WHERE loanType = 'Al-Innah' AND
14    loanApplicationStatus = 'Approved'),
14    (SELECT SUM(shareCapital) FROM Stock),
15    (SELECT SUM(feeCapital) FROM Stock),
16    (SELECT SUM(memberFund) FROM Stock)
17 );

```

Figure 3.1.27 Inserting data into ReportHub table

Month	Year	adminID	bodID	memberNo	total_membershipApplicationStatus	total_loanApplicationStatus	total_shareCapital	total_feeCapital
1	2025	456	123	101	1	1	2000.00	500.00

total_membersFund	applicationID	loanApplicationID	total_loanTypeApplication
3500.00	3001	1001	1

Figure 3.1.28 Data is inserted into ReportHub table

- JOIN Query 1: Combine ReportHub with Membership

```

1 SELECT reporthub.Month, reporthub.Year, membership.applicationStatus
2 FROM reporthub
3 JOIN membership ON reporthub.applicationID = membership.applicationID
4 WHERE membership.membershipStatus = 'Active';

```

Figure 3.1.29 Combine ReportHub with Membership

Month	Year	applicationStatus
1	2025	Success

Figure 3.1.29 Data is inserted

- JOIN Query 2: Combine ReportHub with Loan

```

1 SELECT reporthub.Month, reporthub.Year, loan.loanApplicationID, loan.loanType,
2     loan.loanAmount, loan.loanApplicationStatus
3 FROM ReportHub
4 JOIN loan ON reporthub.loanApplicationID = reporthub.loanApplicationID
5 WHERE loan.loanApplicationStatus = 'Approved';

```

Figure 3.1.30 Combine ReportHub with Loan

Month	Year	loanApplicationID	loanType	loanAmount	loanApplicationStatus
1	2025	1001	AI-Innah	50000.00	Approved

Figure 3.1.31 Data is inserted

- JOIN Query 3: Combine ReportHub with Stock

```

1 SELECT reporthub.Month, reporthub.Year, stock.memberNo, stock.shareCapital,
2    stock.feeCapital, stock.memberFund
3 FROM reporthub
4 JOIN stock ON reporthub.applicationID = stock.applicationID
5 WHERE stock.memberNo = 101;

```

Figure 3.1.32 Combine ReportHub with Stock

Month	Year	memberNo	shareCapital	feeCapital	memberFund
1	2025	101	2000.00	500.00	3500.00

Figure 3.1.33 Data is inserted

- Complex Query: Multi-Table JOIN

```

1 SELECT reporthub.Month, reporthub.Year,
2    membership.name AS MemberName, membership.membershipStatus,
3    loan.loanType, loan.loanAmount, loan.loanApplicationStatus,
4    stock.shareCapital, stock.feeCapital, stock.memberFund
5 FROM reporthub
6 JOIN membership ON reporthub.applicationID = membership.applicationID
7 JOIN loan ON reporthub.loanApplicationID = loan.loanApplicationID
8 JOIN stock ON reporthub.applicationID = stock.applicationID
9 WHERE reporthub.Month = 01 AND reporthub.Year = 2025;

```

Figure 3.1.34 Combine multiple tables

Month	Year	MemberName	membershipStatus	loanType	loanAmount	loanApplicationStatus	shareCapital	feeCapital	memberFund
1	2025	Arjuna Nirmala	Active	AI-Innah	50000.00	Approved	2000.00	500.00	3500.00

Figure 3.1.35 Data is inserted

3.2 INTERFACE IMPLEMENTATION

1) INSERT INTO.... TABLE Membership

PERMOHONAN ANGGOTA KOPERASI

Maklumat Pemohon

A Nama : Arjuna Nirmala Jantina : ☒ L ☐ P No. Anggota : S12345

B No KP : 910101145678 Agama : Islam ▼ No. PF : PF67890

C Taraf Perkahwinan : Berkahwin ▼ Bangsa : Melayu ▼

Alamat Rumah : 123 Jalan Kerinchi

Poskod : 43000 Negeri : Kajang, Selangor

Jawatan & Gred : Grade 1

Alamat Pejabat (Tempat Bertugas) : 456 Corporate Blud

Poskod : 4000 Negeri : Shah Alam, Selangor

No. Telefon Bimbit : 0123456789

No. Telefon/Fax : 031234567 No. Telefon Rumah : 039876543

Gaji Bulanan : RM 5000.00

Seterusnya

helpline : 09-7447088 emel : koperasi_kada@yahoo.com

Figure 3.2.1

PERMOHONAN ANGGOTA KOPERASI

Yuran dan Sumbangan

☒ Jika diterima sebagai anggota, saya bersetuju membayar yuran dan sumbangan bulanan seperti di bawah :

BIL	PERKARA	RM
1.	Yuran Masuk	50
2.	Modal Syer*	2000
3.	Modal Yuran	500
4.	Wang Deposit Anggota	300
5.	Sumbangan Tabung Kebajikan (Al-Abrar)	100
6.	Simpanan Tetap	1000
7.	Lain - lain	--

*Minima modal syer adalah sebanyak RM300.00 dan tidak melebihi 1/3 daripada modal syer koperasi dan hendaklah jelaskan dalam tempoh 8 bulan dari tarikh kelulusan menjadi anggota

Hantar

helpline : 09-7447088 emel : koperasi_kada@yahoo.com

Figure 3.2.2

2) INSERT INTO.... TABLE Loan

The screenshot shows the 'PERMOHONAN PEMBIAYAAN ANGGOTA' (Member Loan Application) form. The form is titled 'Butir-butir Pembiayaan' (Loan Details). It contains the following fields:

- ID Permohonan Pembiayaan :** 1001
- ID Permohonan :** 3001
- Jenis Pembiayaan :**
 - ☐ Al - Bai
 - ☒ Al - Innah
 - ☐ Khas
 - ☐ Karnival Musim Istimewa
 - ☐ Baik Pulih Kenderaan
 - ☐ Road Tax dan Insurans
 - ☐ Al-Qadrol Hassan
 - Lain - Lain :
- Amaun Dipohon :** RM5000.00
- Tempoh Pembiayaan :** 60 bulan
- Ansuran Bulanan :** RM1000.00
- Seterusnya** button

At the bottom, the contact information is provided: helpline : 09-7447088, email : koperasi_kada@yahoo.com.

Figure 3.2.3

The screenshot shows the 'PERMOHONAN PEMBIAYAAN ANGGOTA' (Member Loan Application) form. The form is titled 'Butir-butir Peribadi Pemohon' (Applicant's Personal Details). It contains the following fields:

- Nama :** Arjuna Nirmala
- No KP :** 910101145678
- Tarikh Lahir :** 1991-01-01
- Umur :** 34 tahun
- Nama Bank :** Maybank
- Nombor Akaun Bank :** 123456789
- No KP Penjamin :** 890101-14-5678
- Nama Penjamin :** Zainul Abidin
- ID Ahli Penjamin :** 101
- No Telefon Bimbit Penjamin :** 0123456789
- Resit Pembiayaan :** Receipt_1002.pdf
- Jantina :** ☒ L ☐ P
- Agama :** Islam
- Bangsa :** Malay
- No. Anggota :** S12345
- No. PF :** PF67890
- Jawatan & Gred :** Grade 1
- No. Telefon Bimbit :** 123456789
- No. Telefon/Fax :** 31234567
- Pengakuan :** ☒ filled in
- Pengesahan Majikan :** ☒ filled in
- Hantar** button

At the bottom, the contact information is provided: helpline : 09-7447088, email : koperasi_kada@yahoo.com.

Figure 3.2.4

3) Complex Query: Multi-Table JOIN

KOPERASI KAKITANGAN KADA KELANTAN BERHAD

Laporan Keseluruhan

Profil Koperasi | Statistik Permohonan Pinjaman | Permohonan Ahli | Permohonan Pembiayaan | Penyata Ahli | Rumusan Kewangan | **Laporan Keseluruhan**

LAPORAN TAHUNAN

LAMPIRAN

LAPORAN BULANAN / TAHUNAN

Bulan : 1 Tahun : 2025

Nama Ahli : Arjuna Nirmala lihat ahli sah / lihat seluruh

Jenis Pembiayaan yang Dimohon : Al-Innah

Jumlah Pinjaman : RM 5000.00

Status Permohonan Pembiayaan : Diluluskan

Model Syer : RM2000.00

Model Yuran Tahunan : RM500.00

Tabung Ahli : RM3500.00

helpline : 09-7447088 email : koperasi_kada@yahoo.com

Figure 3.2.5

SECTION D (CONCLUSION AND REFLECTION)

4.1 CONCLUSION

The Board of Directors (BOD) and Administrators are now able to make more informed decisions due to the ReportHub module's efficient centralization of membership and loan application data. It streamlines the generation of monthly and yearly reports, provides summarized financial and operational information, and adheres to Boyce-Codd Normal Form (BCNF) for optimal database design. Additionally, linking reports to specific users enhances traceability through the use of foreign keys.

The module could be enhanced by incorporating more advanced trend analysis visuals, better filters for tailored reporting, and options to export in CSV or Excel formats. Usability and functionality would significantly improve with features such as role-based access controls, automated report scheduling, real-time dashboards, and audit logs. With these enhancements, ReportHub would transform into a more adaptable and powerful tool for decision-making.

4.2 REFLECTION

Description (Pure fact)	To address the inefficiencies in managing loan applications and memberships, our group project focused on developing the KADA Cooperative System. Each member of the group was assigned specific roles within the system, which comprised several modules, including Authentication, Membership, Admin, Approval, Loan, Statement, and ReportHub. I was responsible for the ReportHub module, which gathers operational and financial data to create reports for administrators and the Board of Directors (BOD). The ReportHub module involved building a database that adheres to normalization standards, facilitates monthly and annual reporting, and provides summary data such as membership applications and financial details. My responsibilities included analyzing requirements, developing the module's features, and ensuring its functionality aligned with project objectives. Other team members focused on their respective modules, which encompassed managing member data, processing loan approvals, and creating authentication methods. Collaboration was key, as we shared updates, refined designs based on feedback, and
--------------------------------	--

	ensured that each module integrated seamlessly into the overall system.
Descriptions (Feelings)	I felt a blend of pressure and excitement at that moment. The pressure stemmed from the need to ensure that the ReportHub module met regulations and integrated smoothly with other modules, while the excitement arose from contributing to a meaningful project aimed at enhancing KADA's efficiency. Understanding that the module's functionality was crucial for the decision-making of the Board of Directors (BOD) and administrators, I felt a strong responsibility to deliver high-quality work. Several factors, such as the varying levels of experience among members in database design and coding, influenced the group's activities and behavior. This sometimes led to delays in communication or progress when some individuals required additional assistance or clarification. The challenge was compounded by the need to balance this project with other academic responsibilities. The exercise presented several challenges. Time constraints meant that team members needed to coordinate effectively to meet deadlines. Finding solutions took longer than expected due to a lack of resources, like templates or examples for specific functionalities. Additionally, the dispersed nature of the work made it challenging to maintain clear communication and collaboration, which was essential for smooth integration of the components. Despite these obstacles, the team remained committed and worked together to accomplish their goals.
Evaluation	The project had its ups and downs. On the positive side, it enhanced my teamwork and database design skills, and it was rewarding to see all the pieces fit together. However, I faced pressure from tight deadlines, limited resources, and balancing other responsibilities. Additionally, communication issues and technical challenges, such as integrating modules, added to the difficulties. These factors influenced my behavior by boosting my proactivity and organizational skills. Although the pressure occasionally led to frustration, it motivated me to conduct thorough research and carefully review my work. Given the team's varied areas of expertise and the

	complexity of integration, collaboration and clear communication were crucial. Feedback focused on refining features and aligning modules, which improved the final system and strengthened teamwork.
Analysis	I find this situation worth reflecting on because it highlights the challenges and successes of collaborating on a complex project, especially during the development of the ReportHub module. It underscores the importance of technical skills, teamwork, and effective problem-solving under pressure. This experience taught me the significance of balancing personal responsibilities with open communication. Reflecting on the experience, I realize how my proactive mindset and the success of our group were influenced by my sense of responsibility and the challenges we faced. It highlighted the importance of time management, adaptability, and clear communication. In the future, focusing on clearer job definitions, enhanced planning, and proactive problem-solving will lead to better resource management and collaboration, ensuring smoother integration and more successful projects.
Conclusion	By digitizing loan and membership administration, the KADA Cooperative System successfully addresses the inefficiencies associated with manual operations. The proposed Report Hub module enhances reporting capabilities with features like trend analysis, customizable filters, and export options such as CSV and Excel, ensuring better decision-making support. The database architecture is designed to provide reliable and efficient data management by following normalization standards. This project highlights the importance of user-centric features such as customizable data exports and historical insights. However, there are still unanswered questions regarding the system's scalability, the education of users for the new module, and how user feedback will shape future upgrades.
Self-Criticism	I will focus on conducting thorough user requirements research to anticipate needs such as scalability, insights from historical data, and flexibility in exports to improve my work on related projects. Integrating user feedback early on will be my top priority, and I will ensure that both functionality and performance are

	rigorously tested. Additionally, I will enhance the documentation to facilitate a smoother onboarding process for end users. I plan to seek out resources such as advanced data visualization tools, user training manuals, and expert advice on database optimization for future improvements. By collaborating with colleagues and utilizing feedback methods, we will enhance both the technical aspects and user experience of the system.
--	--